



Predicting disjoint reference: offline and online evidence

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Our Study

Principle B: Pronouns cannot corefer with a local subject.

John_i likes him_{*i/j}.

Why? One factor: general bias for coarguments to be disjoint unless a reflexive is present

Question: Is there a default *disjoint reference bias* for transitive verbs? **Answer:** Yes!

Offline evidence: preference for disjoint interpretations, modulated by verb biases

Online evidence: anticipation of disjoint reference before the object, regardless of bias

Offline Experiments

Do people systematically avoid reflexive interpretations with lexical and nonce verbs with (anti-)reflexivity biases?

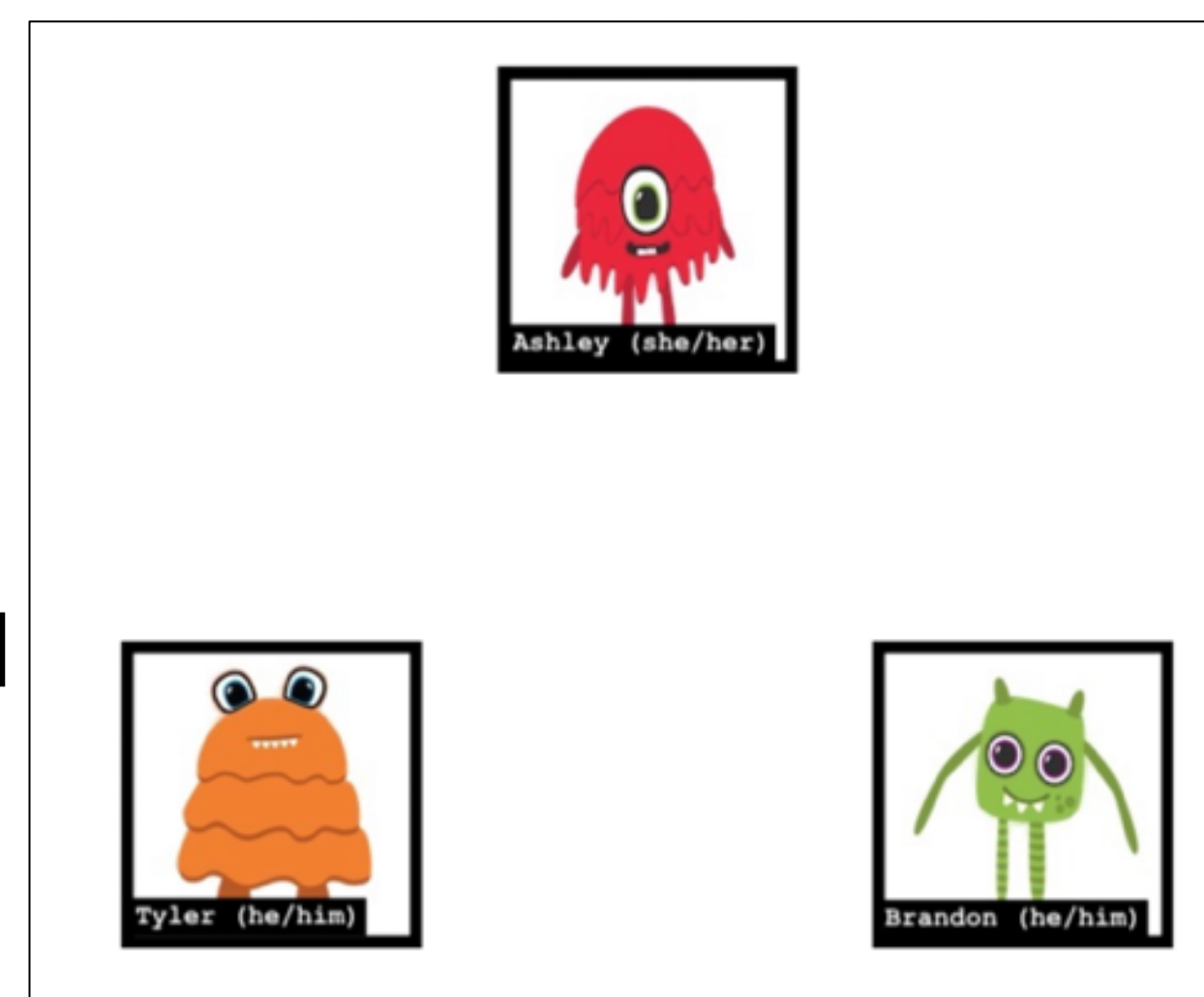
			% of Reflexive Interpretations	
			Exp.1	Exp.2
Wh-Q	Lexical Anti-Refl	Who did Tyler interrupt?	6%	17%
	Lexical Refl	Who did Tyler disgrace?	17%	56%
	Nonce Anti-Refl	Who did Tyler undisgoit?	9%	36%
	Nonce Refl	Who did Tyler intropuct?		71%
RC	Lexical Anti-Refl	Click on the alien who Tyler interrupted.	6%	11%
	Lexical Refl	Click on the alien who Tyler disgraced.	11%	53%
	Nonce Anti-Refl	Click on the alien who Tyler undisgoited.	7%	30%
	Nonce Refl	Click on the alien who Tyler intropucted.		68%

Exp.1 ($N = 18$): only prompts (Wh-Q or RC prompts)

- refl. interpretation in only 9% of the trials,
- (marginally) higher for REFL than ANTI-REFL in lexical verbs ($p = .07$)

Exp.2 ($N = 18$): prompts followed by answers disambiguated by a pronoun in ANTI-REFL or a reflexive in REFL

- refl. interpretation in 45% of the trials,
- higher for REFL than ANTI-REFL in both lexical and nonce verbs (p 's $< .001$)



Example display

Conclusion

- **Default disjoint reference bias** for coarguments of transitive verbs that
 - can be modulated in **offline interpretations**, with both lexical and learned (anti-)reflexivity biases
 - is also seen in **online measures**: people look away from the subject in transitive verbs, no matter what the bias was.
- **Motivation** behind disjoint reference: obviation as a grammatical constraint [4] or statistical generalization

Visual World Experiment

Do listeners anticipate disjoint reference?

Inspired by [1], we test whether they look away from the subject when processing a transitive verb.

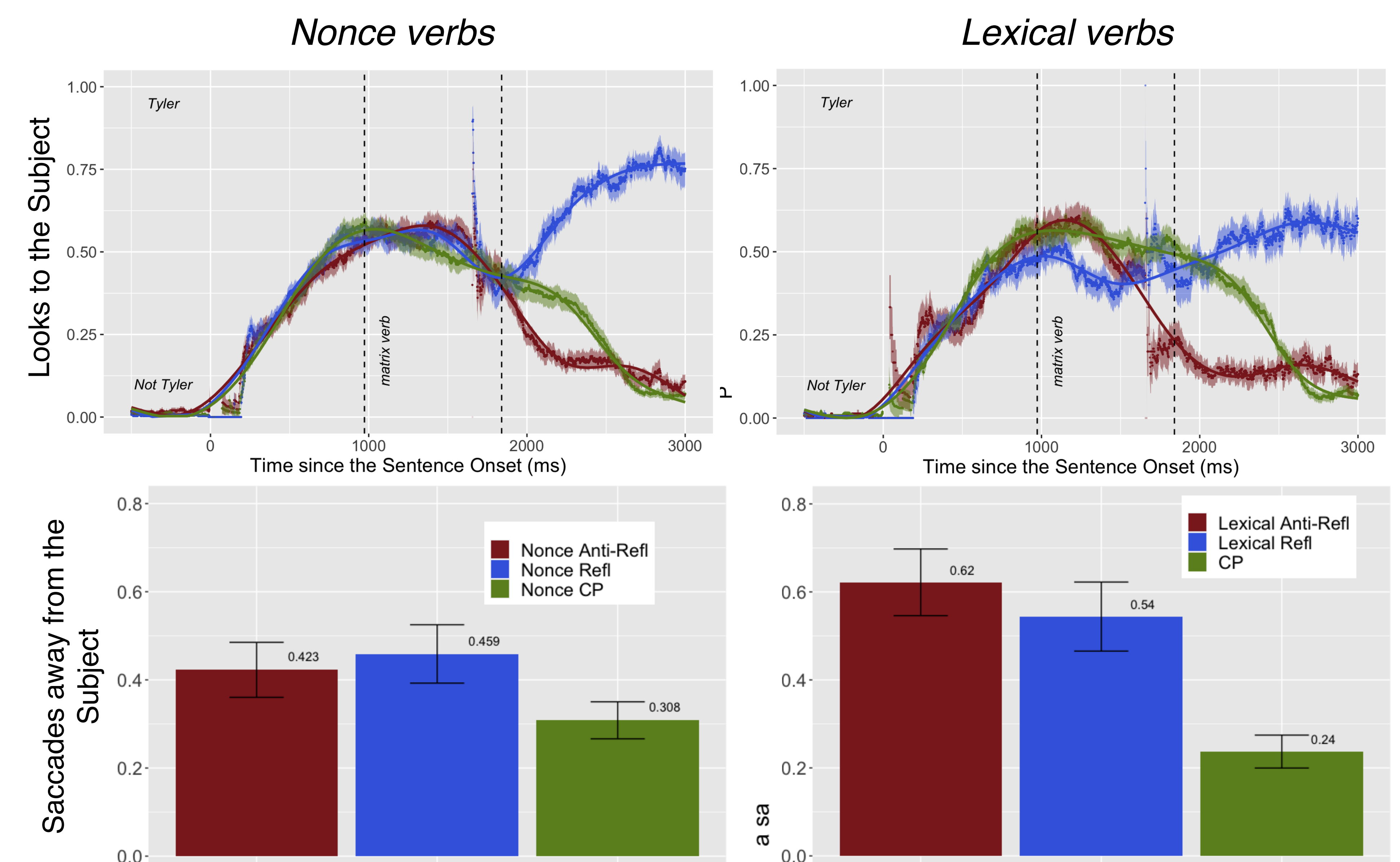
Replicating [2], but with an equal bias for pronouns and reflexives within-experiment.

Exp.3 ($N = 36$): listening to statements with lexical and nonce verbs followed by a pronoun or a reflexive, while looking at the example display

Task: click on the alien that is referred to by the pronoun or the reflexive

A complementizer phrase (CP) condition as a baseline; since there is no direct object after the matrix verb, disjoint reference should not be predicted.

Lexical Anti-Refl	Tyler <u>interrupted</u> her.
Lexical Refl	Tyler <u>disgraced</u> himself.
CP	Tyler <u>deduced</u> that Ashley interrupted her I disgraced herself.
Nonce Anti-Refl	Tyler <u>undisgoited</u> her.
Nonce Refl	Tyler <u>intropucted</u> himself.
Nonce CP	Tyler <u>ippustrated</u> that Ashley undisgoited her I ipportalized herself.



Saccade Analysis [3]: How often do listeners launch a saccade away from the matrix subject within the matrix verb window?

- more saccades away from the matrix subject in ANTI-REFL and REFL than CP condition in both lexical (p 's $< .01$) and nonce verbs (p 's $< .05$).

References: [1] Altmann & Kamide, 1999. [2] Burnsky et al., 2022. [3] Kingston et al., 2016. [4] Kiparsky, 2002.

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